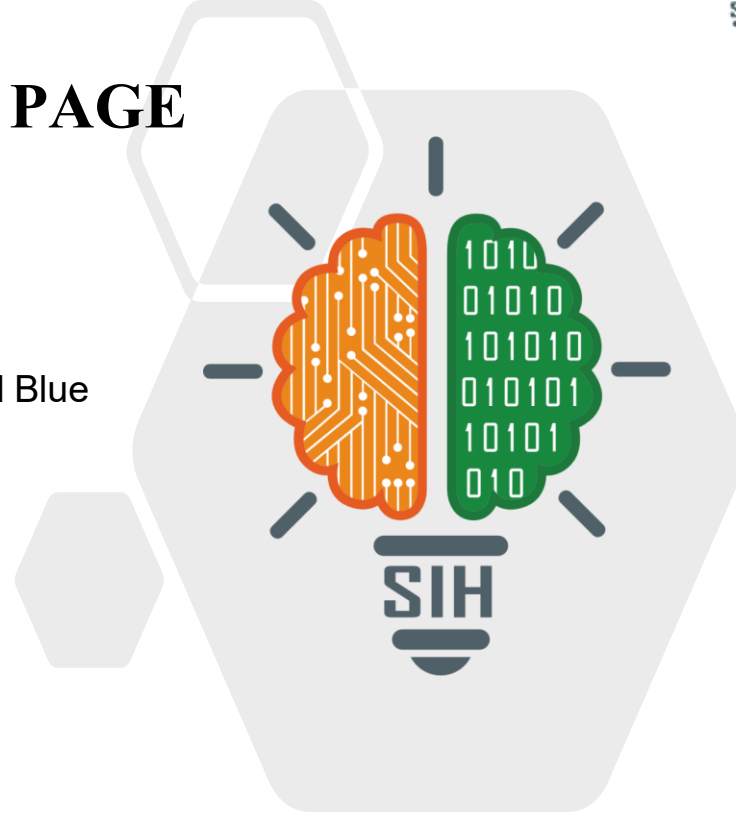


SMART INDIA HACKATHON 2025

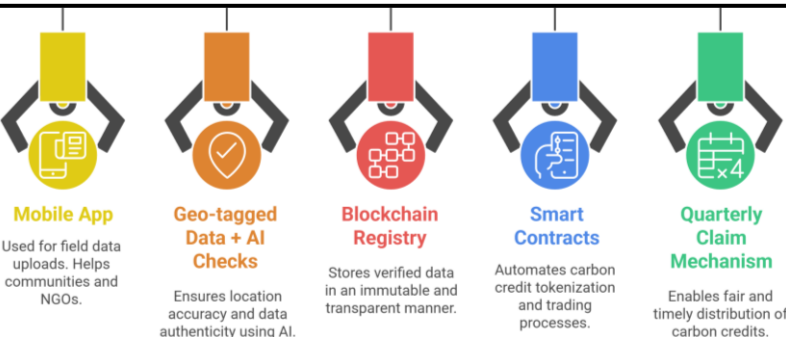


TITLE PAGE

- **Problem Statement ID** – SIH25038
- **Problem Statement Title** - Blockchain-Based Blue Carbon Registry and MRV System
- **Theme**- Clean & Green Technology
- **PS Category** - Software
- **Team ID** - MITADTSW138
- **Team Name**: CarbonX



We propose a **Blockchain-based Blue Carbon Monitoring, Reporting and Verification(MRV) System and Marketplace**, aligned with the initiative of the **Ministry of Earth Sciences (MoES)**, to ensure authentic and transparent reporting of coastal ecosystem restoration and trading of carbon credits. The marketplace will operate as per government policy, where industries are required to purchase credits equivalent to their greenhouse gas emissions, ensuring compliance and credibility.

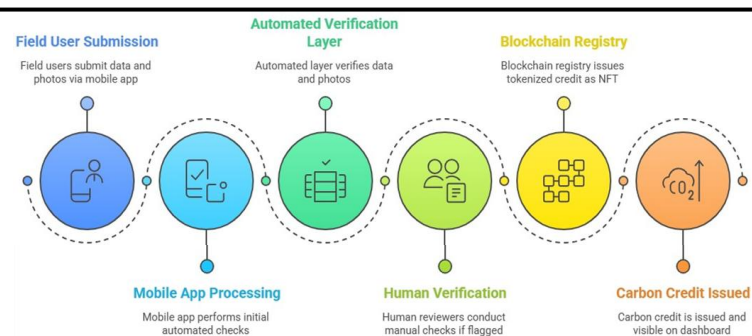


What is the Problem

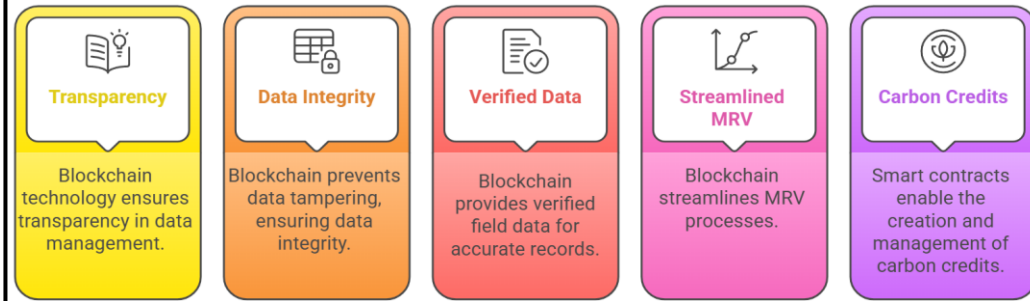


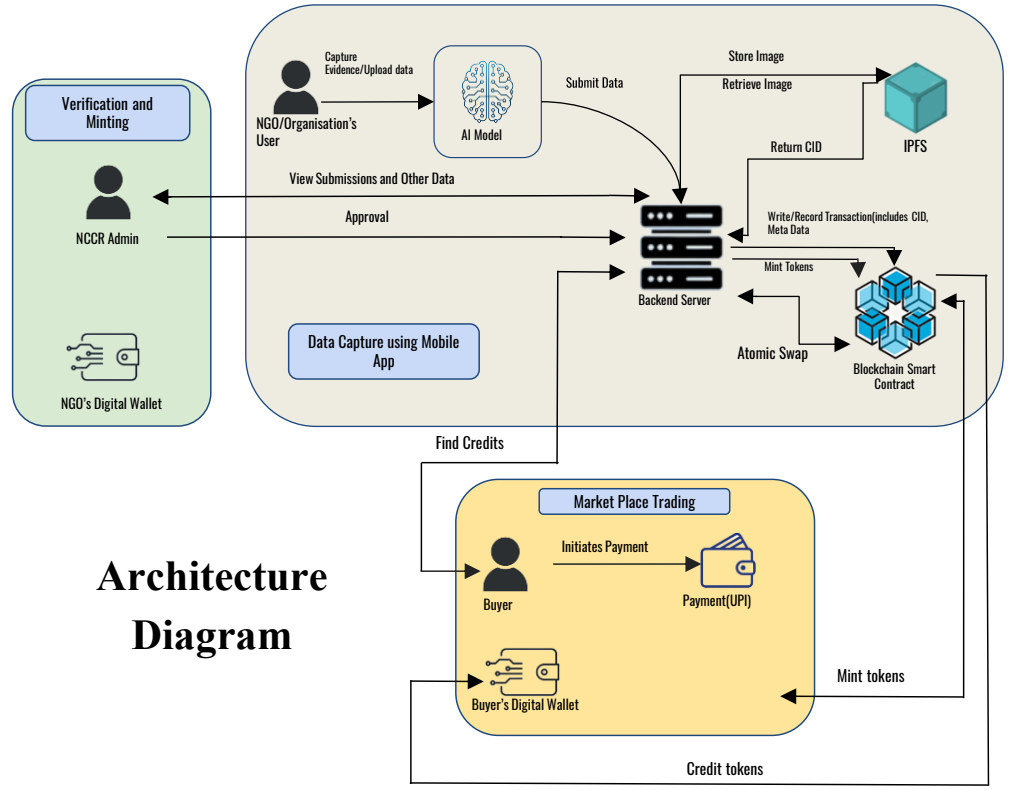
Innovation & Uniqueness

- End-to-End MRV on Blockchain
- NFT-based Carbon Credits
- Inclusive by Design
- AI-verified Photo Capture
- Scalable Roadmap
- Integrated Marketplace

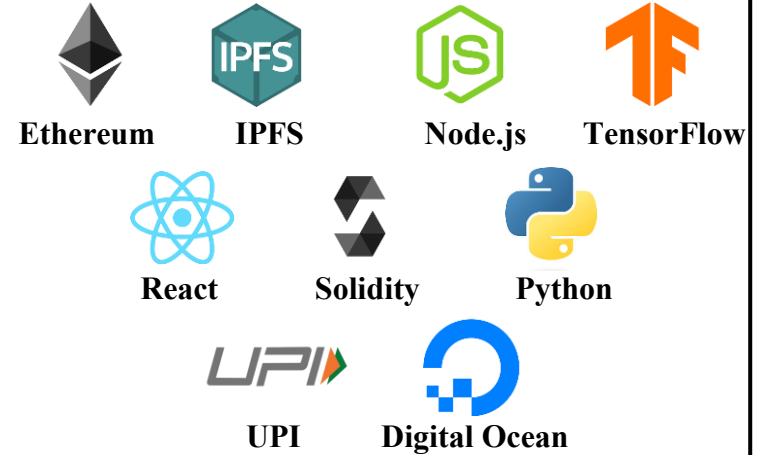


How is the Problem Addressed





Technology-Stack



Prototype Video

<https://drive.google.com/file/d/19XFkHXW12Pp-5qz9aeoqK7BQ5UGWEes3/view?usp=sharing>

Feasibility Analysis



Economic

Self-sustaining business model & growing demand for carbon credits.



Operational

Simplifies MRV, offers easy mobile use, and supports smoother adoption.



Technical

Uses proven tech and efficient design with IPFS for better performance.

Market Validation & Business Model

3.4 million hectares Costal Area of India with Carbon Sequestration $\approx$ 24 million tCO₂/yr

**Registration Fee: ₹100 per hectare
Transaction Tax: 2–3% per credit traded.**

Leading this to a total of 34 Crore Rs. On site registration and 1.92 to 2.88 Crore Rs. taxes over Carbon Credit

Potential Challenges and Risks

Fake or manipulated data submissions (old/gallery/AI-generated photos).

Regulatory uncertainties around India's carbon credit markets.

Limited technical capacity of grassroots users (NGOs, panchayats).

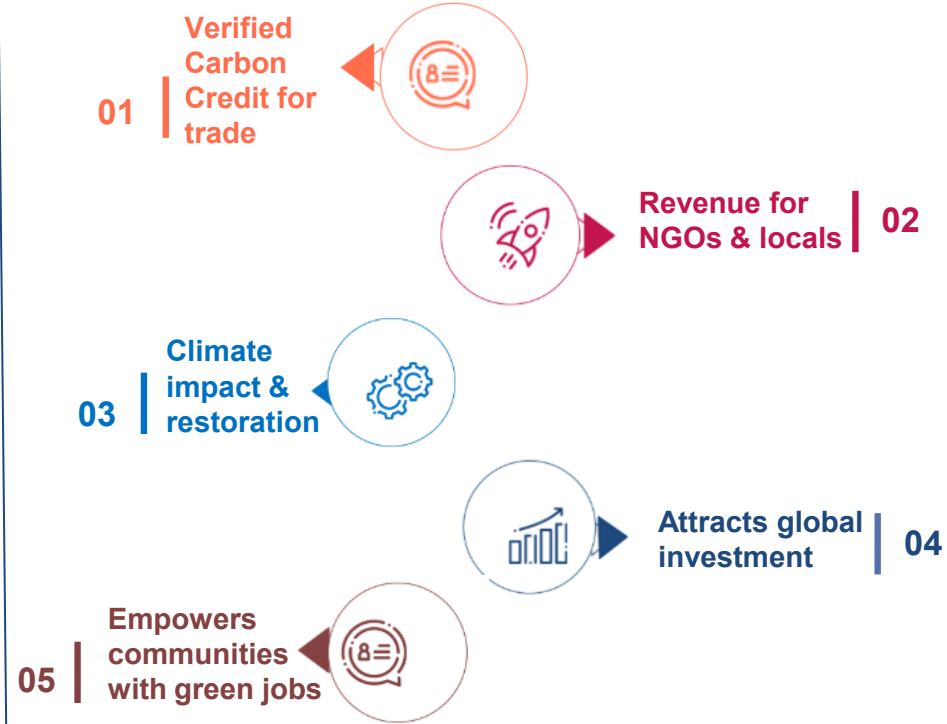
Strategies for Overcoming these Challenges

Enforce **live photo capture** and **AI-based authenticity check + final human review**.

Align with NCCR & Indian CCTS (Carbon Credit Trading Scheme) guidelines.

Provide a simple mobile UI. Training workshops for NGOs/panchayats.

IMPACT AND BENEFITS



Reference / Title	Key Finding & Relevance to Our Project	Source / Link
The Missing Piece: Blue Carbon in India's Voluntary Carbon Market (2025)	India's blue carbon potential is locked away due to a lack of standardized verification and direct financial incentives.	Source/Link
PARLIAMENT QUESTION: BLUE CARBON PROJECT (2025)	While national blue carbon initiatives like MISHTI exist, there is a recognized gap in specific guidelines - need for a standardized verification system	Source/Link
The rise and flows of blue carbon credits advance global climate and biodiversity goals (2025)	The Blue Carbon Project (BCP) market is expanding rapidly, with a major surge in new projects in recent years.	Source/Link
Exploring blue carbon research in India: current landscape and future avenues (2025)	Mangroves, seagrasses, and salt marshes are blue carbon ecosystems that sequester carbon far more efficiently than land-based forests, making them vital for mitigating climate change	Source/Link
India's Carbon Market Potential (2025)	To succeed in the EU carbon market, India needs a transparent registry to ensure quality credits and prevent double-counting.	Source/Link
Indian Carbon Credit Trading Scheme	India's new Carbon Credit Trading Scheme (CCTS), help industries reduce emissions intensity and support the country's climate goals.	Source/Link
Applying blockchain technology for building energy performance measurement, reporting, and verification (MRV) and the carbon credit market (2021)	Blockchain provides a trustworthy, automated framework for MRV and carbon trading, ensuring both the immutability of its registry and complete transparency.	Source/Link